Multiple Crossover Phenomena and Scale Hopping in Two Dimensions

Source-backed arXiv sample

1 Extracted Lists

The list environments below are extracted from arXiv LaTeX source and wrapped for pdfLaTeX validation.

1. Positions of the nearly marginal thermal operators in the Kac table of a unitary minimal model M_p for $p \gg 1$. The operators $\phi_{(n,n+2)}$ are relevant, while the operators $\phi_{(n,n-2)}$ and $\tilde{\phi}_{(n,n)}$ are irrelevant.

2. Special solutions of the RG equations in the vicinity of $M_{p,p-1}$ (schematic). (a) A trajectory in $\mathcal{R}_p$. (b) A trajectory in $\mathcal{C}_{p-1}$. (c) A redundant trajectory.

3. Self-conjugate trajectories in the vicinity of the fixed point M_p (schematic). (a) A trajectory in $M_{p+1,p-1}$. (b) A self-similar trajectory.

4. The C-function $C(\tau, \tau_0)$ of the unique self-similar trajectory $\check{u}_p^i(\tau, s(\tau_0))$: a step in the staircase pattern for $\tau_0 = 3.2$, 3.6, 4.0 and 4.4 (solid lines). For larger values of τ_0, the steps get more pronounced as the solutions tend towards the limit trajectory $M_{p,p-1}$ (long-dashed line).

5. (a) The RG flow in the vicinity of M_p. A self-similar trajectory of regime 1 (solid line) visits all fixed points, trajectories in regime 2 or 3 (long-dashed lines) visit two fixed points, and a trajectory in regime 4 (short-dashed line) visits only one fixed point. (b) The resulting phase diagram in the $(t_p, \bar{t}_p)$-plane.